Time Evolution

In quantum mechanics, the time evolution of a closed system is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle . \tag{1}$$

For a time-independent Hamiltonian $\hat{H}(t) = \hat{H}$, we can readily solve this equation to find the time evolution of the state vector $|\psi(t)\rangle$:

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t} |\psi(0)\rangle , \tag{2}$$

motivating us to define the time-evolution operator

$$\hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}t}. \tag{3}$$

If the Hamiltonian is time-dependent, the time evolution operator is given by the time-ordered exponential:

$$\hat{U}(t) = \mathcal{T} \exp\left(-\frac{i}{\hbar}\int_0^t \hat{H}(t')dt'\right), \tag{4}$$

where $\mathcal{T}$ denotes the time-ordering operator. Time ordering is necessary because, in general, the Hamiltonian at different times may not commute, $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$ for $t_1 \neq t_2$ (we interpret this case a little more in Section A). So, the time evolution operator must account for the order in which the Hamiltonians are applied. If the Hamiltonian is time-dependent, but its evolution does not care about the order of things (i.e., if $[\hat{H}(t_1), \hat{H}(t_2)] = 0$ for all t_1, t_2), then time ordering is irrelevant and we get

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\int_0^t \hat{H}(t')dt'\right). \tag{5}$$

> The result in Eq. (4) is given in S&N Chapter 2.1.2 and the derivation deferred to Chapter 5. You can find a sketch of the derivation in Section B, where we introduce the *Dyson Series*. The evaluation of time-ordered exponentials can be complicated and may require approximate methods to get to a solution. We will return to this issue in time-dependent perturbation theory, where the Dyson series will come back to aid (or bite?) us.

For a hermitian Hamiltonian, this operator is unitary, ensuring that the total probability is conserved over time. The state of the system at time t can be obtained by applying the time evolution operator to the initial state $|\psi(0)\rangle$:

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle . \tag{6}$$

The time evolution operator satisfies the following properties:

- Unitarity: $\hat{U}^\dagger(t)\hat{U}(t) = \hat{U}(t)\hat{U}^\dagger(t) = \hat{I}$.
- Composition: $\hat{U}(t_2 + t_1) = \hat{U}(t_2)\hat{U}(t_1)$.
- Initial condition: $\hat{U}(0) = \hat{I}$.

The expectation value of an observable $\hat{A}$ at time t is given by

$$\boxed{\langle\hat{A}\rangle_t = \langle\psi(0)|\hat{U}^\dagger(t)\hat{A}\hat{U}(t)|\psi(0)\rangle.} \tag{7}$$

Schrödinger versus Heisenberg Picture

From Eq. (7), we can see that there is an ambiguity in how we attribute the time dependence in quantum mechanics. One may easily interpret the expression as the time evolution operators acting on the states $|\psi(0)\rangle$, and then taking the expectation value of a fixed operator $\hat{A}$. But we may also equivalently interpret the expression as the time evolution operators acting on the operator $\hat{A}$, while the states remain fixed at their initial values. This leads to two different but equivalent pictures of quantum mechanics:

- **Schrödinger Picture:** States evolve in time, operators are fixed,

$$\boxed{|\psi(t)\rangle = \hat{U}(t)\,|\psi(0)\rangle, \quad \hat{A}(t) = \hat{A}}. \tag{8}$$

- **Heisenberg Picture:**Operators evolve in time, states are fixed,

$$\boxed{|\psi(t)\rangle = |\psi(0)\rangle, \quad \hat{A}(t) = \hat{U}^\dagger(t)\hat{A}\hat{U}(t)}. \tag{9}$$

Which picture to choose from is often a matter of convenience. Schrödinger's picture is more intuitive when asking questions about the states themselves, such as what a wave packet does as it travels in space. Heisenberg's picture is often more convenient for dealing with time-independent states and focusing on the dynamics of observables, such as in time-dependent Hamiltonians. When studying time-dependent perturbation theory, we will make use of a hybrid version of these: the interaction picture. What is important is that the connection to physical observables, such as expectation values, remains the same in all prescriptions.

By difderentiating Eq. (9), show that the Equation of motion in the Heisenberg picture is given by

$$\frac{d\hat{A}(t)}{dt} = \frac{1}{i\hbar}\left[\hat{A}(t), \hat{H}_H(t)\right] + \left(\frac{\partial \hat{A}}{\partial t}\right)_{\text{explicit}}, \tag{10}$$

where $\hat{H}_H(t) = \hat{U}^\dagger(t)\hat{H}\hat{U}(t)$ is the Hamiltonian in the Heisenberg picture. Note that the additional term on the right-hand side is there to capture any explicit time dependence of $\hat{A}$ (not common). If $\hat{H}$ is time-independent, then $\hat{H}_H(t) = \hat{H}$. Note the important difference between $\hat{H}_H$ and $\hat{H}(t)$: one is the Hamiltonian in the Heisenberg picture, the other is our original Hamiltonian as a function of time in (which we used assuming the Schrödinger picture, so you may also see $\hat{H}(t) = \hat{H}_S(t)$).

Propagators

The application of the time evolution operator can also be expressed in terms of *propagators*. These are Green's functions (see notes on Green's functions) that describe how the wave function propagates from one point in space and time to another. In the position basis, the propagator $K(x_1, t_1; x_0, t_0)$ is defined as

$$K(x_1, t_1; x_0, t_0) = \langle x_1|\,\hat{U}(t_1, t_0)\,|x_0\rangle\,. \tag{11}$$

In the Schrödinger picture, we may be even as bold as to write

$$K(x_1, t_1; x_0, t_0) = \langle x_1, t_1|x_0, t_0\rangle\,, \tag{12}$$

where $|x_1, t_1\rangle = \hat{U}(t_1, t_0)\,|x_1, t_0\rangle$. Note what we have here: the propagator is the matrix element (aka, the transition amplitude, a type of expectation value) of the time evolution operator between two position eigenstates. In other words, we can say it is the amplitude for a particle to propagate from position x_0 at time t_0 to position x_1 at time t_1.

How is this useful? Well, if we know the propagator, we can determine how any initial wavefunction $\psi(x_0, t_0)$ evolves in time, and that is a big deal.

Specifically, show that the wavefunction at time t_1 can be expressed in terms of the propagator and the initial wavefunction at time t_0 as

$$\psi(x_1, t_1) = \int dx_0\, K(x_1, t_1; x_0, t_0)\psi(x_0, t_0), \tag{13}$$

where $\psi(x, t) = \langle x|\psi(t)\rangle$. In addition, prove the following properties of the propagator:

- **Initial condition**: At equal times, the propagator reduces to the Dirac delta function

$$K(x_1, t; x_0, t) = \delta(x_1 - x_0). \tag{14}$$

- **Composition**: The propagator satisfies the composition property

$$K(x_2, t_2; x_0, t_0) = \int dx_1\, K(x_2, t_2; x_1, t_1)K(x_1, t_1; x_0, t_0), \tag{15}$$

  for any intermediate time t_1.

Great, if we know the propagator, we can find how any wavefunction evolves in time. But how do we find the propagator itself? It turns out that it obeys its own Schrödinger equation[1]:

$$i\hbar\frac{\partial}{\partial t_1}K(x_1, t_1; x_0, t_0) = \hat{H}(x_1, t_1)K(x_1, t_1; x_0, t_0), \tag{16}$$

with the initial condition $K(x_1, t_1; x_0, t_1) = \delta(x_1 - x_0)$. Solving this equation for the propagator can be challenging, but for simple systems like the free particle or the harmonic oscillator, explicit expressions for the propagator can be found. But most importantly, the propagator is a tool that is very useful to understand the path integral formulation of quantum mechanics and to solve many problems in quantum dynamics (as we will see later in this course). Ultimately, it is also a language that is fundamental to quantum field theory.

Free Particle

Let us revisit the free particle Hamiltonian in three dimensions:

$$\hat{H} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2. \tag{17}$$

The time evolution operator with $t_0 = 0$ for the free particle is given by:

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m}t}. \tag{18}$$

To find the propagator in the position basis, we compute

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \langle \mathbf{x}_1|\, \hat{U}(t)\, |\mathbf{x}_0\rangle = \langle \mathbf{x}_1|\, e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m}t}\, |\mathbf{x}_0\rangle\,. \tag{19}$$

[1] You can show this by differentiating the definition of the propagator with respect to t_1 and using $\partial U(t_1, t_0)/\partial t_1 = -\frac{i}{\hbar}\hat{H}(t_1)U(t_1, t_0)$.

Inserting a complete set of momentum eigenstates, we have

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \int d^3p \, \langle \mathbf{x}_1 | \, e^{-\frac{i}{\hbar}E_p t} \, |\mathbf{p}\rangle \, \langle \mathbf{p}|\mathbf{x}_0\rangle \,, \tag{20}$$

where we defined the energy of the free particle as $E_p = \frac{p^2}{2m}$. Using the property $\langle \mathbf{x}| \, \mathbf{p} = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{x}}$, we find

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \frac{1}{(2\pi\hbar)^3} \int d^3p \, e^{-\frac{i}{\hbar}E_p t} e^{\frac{i}{\hbar}\mathbf{p}\cdot(\mathbf{x}_1 - \mathbf{x}_0)}. \tag{21}$$

This integral is a relatively straightforward Gaussian integral and can be evaluated to yield

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \left(\frac{m}{2\pi i\hbar t}\right)^{3/2} \exp\left(\frac{im|\mathbf{x}_1 - \mathbf{x}_0|^2}{2\hbar t}\right). \tag{22}$$

This propagator describes how a free particle's wavefunction evolves over time, spreading out as it propagates through space.

As expected, Eq. (22) satisfies the initial condition (with $t \to 0$, $K(\mathbf{x}_1, t; \mathbf{x}_0, 0) \to \delta(\mathbf{x}_1 - \mathbf{x}_0)$) and composition properties of the propagator.

> Fourier transform the propagator in Eq. (22) back to momentum space to show that
>
> $$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \langle \mathbf{p}_1 | \, \hat{U}(t) \, |\mathbf{p}_0\rangle = \delta(\mathbf{p}_1 - \mathbf{p}_0) e^{-\frac{i}{\hbar}E_{p_0} t}. \tag{23}$$
>
> This shows that in momentum space, the free particle propagator simply adds a phase factor to each momentum eigenstate, a consequence of the fact that momentum eigenstates are energy eigenstates for the free particle.

A Non-Commuting Hamiltonians

To understand the case where $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$, we will need two basic facts:

1) the expectation value of an operator that does not commute with the Hamiltonian is not conserved in time,

2) if two operators commute, they share a common set of eigenstates. In other words, they can be simultaneously diagonalized.

Prove that $\frac{d}{dt}\left\langle \hat{A} \right\rangle = \frac{i}{\hbar}\left\langle [\hat{H}, \hat{A}] \right\rangle + \frac{\partial}{\partial t}\left\langle \hat{A} \right\rangle$, where the last term is present if $\hat{A}$ has explicit time dependence.

Prove that if $[\hat{A}, \hat{B}] = 0$, then the eigenstates of $\hat{A}$ are also eigenstates of $\hat{B}$, that is, if $\hat{A}\left|a\right\rangle = a\left|a\right\rangle$, then $\hat{B}\left|a\right\rangle = b\left|a\right\rangle$ for some eigenvalue b. It need not be true that $a = b$. You can assume that the eigenvalues are non-degenerate for simplicity.

Therefore, if the Hamiltonians at different times do not commute, it means that energy is not conserved in our system (due to non-equilibrium processes, contact with an external bath, or some external varying magnetic field, for example). In addition, the energy eigenstates of the system may be changing with time (in the Schrödinger picture). Note that the energy eigenstates are also be changing when $[\hat{H}(t_1), \hat{H}(t_2)] = 0$ (including in the time-independent case), but only by a phase factor. The point is that in the most general case, the energy eigenstates at a future time may not just be a phase-evolved basis, but rather, "directions" may change too. If we were to insist on using the initial eigenbasis, we simply cannot guarantee that once we start evolving it in time, that the new energy eigenbasis is aligned with the initial one (this is equivalent to saying that a vector in the new eigenbasis evolves to be a mix of initial ones).

A.1 Single qubit example

To see this in action, let us consider a two-level system (qubit) with a time-dependent Hamiltonian given by

$$\hat{H}(t) = \frac{\hbar\omega_0}{2}\hat{\sigma}_z + V(t)\hat{\sigma}_x, \tag{24}$$

where $\hat{\sigma}_x$ and $\hat{\sigma}_z$ are the Pauli matrices, ω_0 is a constant frequency, and $V(t)$ is a time-dependent potential. The Pauli matrices do not commute, i.e., $[\hat{\sigma}_x, \hat{\sigma}_z] = -2i\hat{\sigma}_y$ (equivalently $[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y$), so the Hamiltonians at different times do not commute either unless $V(t_1) = V(t_2)$ (e.g. V is constant):

$$[\hat{H}(t_1), \hat{H}(t_2)] = \frac{\hbar\omega_0}{2}(V(t_2) - V(t_1))[\hat{\sigma}_z, \hat{\sigma}_x] = i\hbar\omega_0(V(t_2) - V(t_1))\hat{\sigma}_y \neq 0, \tag{25}$$

The energy change from time t_1 to t_2 can be computed as

$$\Delta E = \langle \psi(t_2)|\hat{H}(t_2)|\psi(t_2)\rangle - \langle \psi(t_1)|\hat{H}(t_1)|\psi(t_1)\rangle\,, \tag{26}$$

which is in general non-zero due to the time dependence of $V(t)$. We can estimate this difference for a small step in time $\Delta t = t_2 - t_1$ and writing $\left|\psi(t_2)\right\rangle = \hat{U}(t_2, t_1)\left|\psi(t_1)\right\rangle \simeq \left(\hat{I} - \frac{i}{\hbar}\hat{H}(t_1)\Delta t\right)\left|\psi(t_1)\right\rangle$. In that case,

$$\begin{aligned}
\Delta E &\simeq \langle \psi(t_1)|\hat{H}(t_2) - \hat{H}(t_1)|\psi(t_1)\rangle + \frac{i}{\hbar}\Delta t\,\langle \psi(t_1)|[\hat{H}(t_1), \hat{H}(t_2)]|\psi(t_1)\rangle + \mathcal{O}(\Delta t^3) &(27)\\
&= \langle \psi(t_1)|(V(t_2) - V(t_1))\left(\hat{\sigma}_x - \omega_0\Delta t\hat{\sigma}_y\right)|\psi(t_1)\rangle + \mathcal{O}(\Delta t^3). &(28)
\end{aligned}$$

Let us consider a specific system at t_1 with $|\psi(t_1)\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, where $|0\rangle$ and $|1\rangle$ are the eigenstates of $\hat{\sigma}_z$ with eigenvalues $+1$ and -1, respectively. At time t_1, the expectation value of the Hamiltonian is

$$\langle\psi(t_1)|\hat{H}(t_1)|\psi(t_1)\rangle = V(t_1). \tag{29}$$

At time $t_2 = t_1 + \Delta t$, the expectation value of the Hamiltonian is

$$\langle\psi(t_2)|\hat{H}(t_2)|\psi(t_2)\rangle \simeq V(t_2) - \omega_0 \Delta t \left(V(t_2) - V(t_1)\right). \tag{30}$$

Therefore, the energy change over the time interval Δt is

$$\Delta E \simeq \left(V(t_2) - V(t_1)\right)\left(1 - \omega_0 \Delta t\right). \tag{31}$$

B Time-Ordered Evolution and the Dyson Series

In what follows, we give a brief derivation of Eq. (4) (well... as brief as I could make it). The essence of the derivation is based on the fact that for an infinitesimal time step dt, the unitary time evolution is

$$\hat{U}(t+dt,t) = \hat{I} - \frac{i}{\hbar}\hat{H}(t)dt + \mathcal{O}(dt^2).$$

with $H(t)$ evaluated at time t. We will not assume anything about $\hat{H}(t)$ other than that it is hermitian. In particular, we do not assume that $\hat{H}(t_1)\hat{H}(t_2) = \hat{H}(t_2)\hat{H}(t_1)$ for $t_1 \neq t_2$ (that is, the Hamiltonians at different times do not necessarily commute).

Knowing the form of the infinitesimal time evolution operator, we can build up the full time evolution operator by composing many infinitesimal steps:

$$\hat{U}(t = t_N, t_0) = \hat{U}(t_N, t_{N-1})\,\hat{U}(t_{N-1}, t_{N-2}) \cdots \hat{U}(t_1, t_0). \tag{32}$$

It is important that the earliest time evolutions appear on the rightmost side of this product. Using our infinitesimal form for $U(t+dt,t)$ for uniform steps $\Delta t = t_{k+1} - t_k$, we have

$$\hat{U}(t = t_N, t_0) = \left(\hat{I} - \frac{i}{\hbar}\,\hat{H}(t_{N-1})\,\Delta t + \ldots\right) \ldots \left(\hat{I} - \frac{i}{\hbar}\,\hat{H}(t_1)\,\Delta t + \ldots\right)\left(\hat{I} - \frac{i}{\hbar}\,\hat{H}(t_0)\,\Delta t + \ldots\right), \tag{33}$$

Multiplying these out gives us terms of several orders in Δt (the lowest order being $(\Delta t)^0$ and highest order being $(\Delta t)^N$). Let us start collecting terms order by order and take the continuum limit $N \to \infty$ (with $N \times \Delta t$ fixed), also order by order. For convenience, let us define:

$$t_N \overset{N\to\infty}{\longrightarrow} t_\infty = t. \tag{34}$$

Order $(\Delta t)^0$**:** There is only one such term, which is simply the identity operator:

$$\hat{I} \quad \overset{N\to\infty}{\longrightarrow} \quad \hat{I}. \tag{35}$$

Order $(\Delta t)^1$**:** There are N such terms, each coming from picking one factor of $-\frac{i}{\hbar}\hat{H}(t_k)\Delta t$ from one of the N factors in Eq. (33), and $\hat{I}$ from all the others. So, the sum of all order $(\Delta t)^1$ terms is

$$-\frac{i}{\hbar}\sum_{k=0}^{N-1}\hat{H}(t_k)\,\Delta t \quad \overset{N\to\infty}{\longrightarrow} \quad -\frac{i}{\hbar}\int_{t_0}^{t} dt_1\,\hat{H}(t_1). \tag{36}$$

Note that t_1 is a dummy integration variable now (not to be confused with the discrete time steps t_k). For a given t, the result above is a simple integral from t_0 to t. (Note that k is counting the number of intervals, so we count it from 0 to $N-1$ here.)

Order $(\Delta t)^2$**:** There are two kinds of terms at this order. First and most important are those from multiplying $\mathcal{O}(\Delta t)$ terms from different $\hat{U}$ factors in Eq. (33). We find a total of

$$\binom{N}{2} = \frac{N(N-1)}{2}, \tag{37}$$

of such terms, which we can write as

$$\left(-\frac{i}{\hbar}\right)^2 \sum_{k_2 > k_1 = 0}^{N-1} \hat{H}(t_{k_2})\hat{H}(t_{k_1})\,(\Delta t)^2 \quad \overset{N\to\infty}{\longrightarrow} \quad \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1\,\hat{H}(t_2)\hat{H}(t_1), \tag{38}$$

Note that each Hamiltonian factor must appear at the same time or at decreasing time orders since the earliest time evolutions appear on the rightmost side of the product in Eq. (33). For a fixed time t, we are integrating over all possible triangular regions in the t_1-t_2 plane such that $t_0 < t_1 < t_2 < t$.

The second kind of contribution at $\mathcal{O}(\Delta t)^2$ comes from picking a term of $\mathcal{O}(\Delta t)^2$ from a single $\hat{U}$ factor. Clearly, there must be only a total of N of such terms, which we can write as

$$\frac{1}{2!}\left(-\frac{i}{\hbar}\right)^2 \sum_{k=0}^{N-1} \hat{H}(t_k)\hat{H}(t_k)\,(\Delta t)^2 = \frac{1}{2!}\left(-\frac{i}{\hbar}\right)^2 \sum_{k_1,k_2=0}^{N-1} (\Delta t\,\hat{H}(t_{k_1}))(\Delta t\,\hat{H}(t_{k_2})\delta_{k_1k_2} \overset{N\to\infty}{\longrightarrow} 0. \tag{39}$$

One might be tempted to say that the continuum limit ought to recover something like a double integral over t_1 and t_2 with a delta function enforcing $t_1 = t_2$, something like

$$\int_{t_0}^{t} dt_2 \int_{t_0}^{t} dt_1\,\hat{H}(t_2)\hat{H}(t_1)\,\delta(t_2-t_1).$$

However, this is not the case. You can convince yourself by just noting that the delta function has units of 1/time, while the Kronecker delta $\delta_{k_1k_2}$ is dimensionless. The correct continuum limit that does recover the delta function would be

$$\frac{\delta_{k_1k_2}}{\Delta t} \to \delta(t_{k_1} - t_{k_2}), \tag{40}$$

which is clearly not what we have in Eq. (39). Therefore, this second kind of contribution at order $(\Delta t)^2$ actually vanishes in the continuum limit. It corresponds to the very edge (the hypotenuse) of the triangular integration region, which has measure zero in the two-dimensional integral. Therefore, we can safely ignore it.

Order $(\Delta t)^n$**:** The pattern established above continues for higher orders in Δt. The leading number of contributions comes from picking n different Δt factors from n different $\hat{U}$ factors in Eq. (33), which gives us a total number of terms equal to

$$\binom{N}{n} = \frac{N!}{n!(N-n)!} \quad \overset{N\to\infty}{\longrightarrow} \quad \frac{N^n}{n!}. \tag{41}$$

The sum of all such leading terms is

$$\begin{aligned}&\left(-\frac{i}{\hbar}\right)^n \sum_{k_n>\cdots>k_1=0}^{N-1} \hat{H}(t_{k_n})\hat{H}(t_{k_{n-1}})\cdots\hat{H}(t_{k_1})\,(\Delta t)^n \\ &\quad\overset{N\to\infty}{\longrightarrow} \quad \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t} dt_n \int_{t_0}^{t_n} dt_{n-1}\cdots\int_{t_0}^{t_2} dt_1\,\hat{H}(t_n)\hat{H}(t_{n-1})\cdots\hat{H}(t_1).\end{aligned} \tag{42}$$

We can say that for a fixed time t, we are integrating over all possible n-dimensional "polytopes" in the space of t_1-t_2-$\cdots$-t_n defined by $t_0 < t_1 < t_2 < \cdots < t_n < t$.

Just as before, this does not capture the edges of the n-dimensional integration region where two or more of the time integration variables are equal ($t_1 = t_2$ or $t_1 = t_2 = \cdots = t_n$, etc), but again, we need not worry about these since these edge regions have measure zero in the n-dimensional integral. In fact, by inspection, one may guess that there should be a total of

$$\sum_{\ell=1}^{n-1}\binom{N}{\ell} \quad \overset{N\to\infty}{\longrightarrow} \quad \frac{N^{(n-1)}}{(n-1)!} + \frac{N^{(n-2)}}{(n-2)!} + \cdots + N \tag{43}$$

such "edge" contributions coming from picking $\mathcal{O}(\Delta t)^{n-\ell}$ terms from ℓ different $\hat{U}$ factors and $\mathcal{O}(\Delta t)^{\ell}$ terms from a single $\hat{U}$ factor. Clearly, these cannot compete with the leading $N^n/n!$

behavior, and hence vanish in the continuum limit.

Let us now collect all orders to find the *Dyson series*:

$$\boxed{\begin{aligned}\hat{U}(t,t_0) = \hat{I} + \left(-\frac{i}{\hbar}\right)\int_{t_0}^{t} dt_1\, \hat{H}(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1\, \hat{H}(t_2)\hat{H}(t_1) + \cdots \\ + \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t} dt_n \int_{t_0}^{t_n} dt_{n-1} \cdots \int_{t_0}^{t_2} dt_1\, \hat{H}(t_n)\hat{H}(t_{n-1})\cdots\hat{H}(t_1) + \cdots .\end{aligned}} \tag{44}$$

This looks quite daunting, so we introduce an important piece of notation: the *time-ordering operator* $\mathcal{T}$. The definition of time ordering is formally very straightforward:

$$\mathcal{T}\left\{\hat{H}(t_1)\hat{H}(t_2)\right\} = \theta(t_1-t_2)\hat{H}(t_1)\hat{H}(t_2) + \theta(t_2-t_1)\hat{H}(t_2)\hat{H}(t_1) = \begin{cases}\hat{H}(t_1)\hat{H}(t_2), & \text{if } t_1 > t_2, \\ \hat{H}(t_2)\hat{H}(t_1), & \text{if } t_2 > t_1.\end{cases} \tag{45}$$

where $\theta(t)$ is the Heaviside step function defined as $\theta(t) = 1$ for $t > 0$ and $\theta(t) = 0$ for $t < 0$. The $t = 0$ case is again irrelevant, but may be set to $1/2$ to show how the time-ordering operator handles equal times symmetrically. A more precise and general definition is:

$$\mathcal{T}\left\{\hat{H}(t_1)\cdots\hat{H}(t_n)\right\} = \sum_{\pi\in S_n} \hat{H}(t_{\pi(1)})\cdots\hat{H}(t_{\pi(n)})\theta(t_{\pi(1)}-t_{\pi(2)})\cdots\theta(t_{\pi(n-1)}-t_{\pi(n)}), \tag{46}$$

where the sum is over all permutations π in the symmetric group S_n on the set $\{1,2,\ldots,n\}$ (fancy way of saying: all possible arrangements of the numbers $1,2,\ldots,n$).

The ordering of the limits in the integral is very awkward, but by applying the time-ordering operator, one may symmetrize each nested integral and let $\mathcal{T}$ take care of the ordering for us:

$$\int_{t_0}^{t} dt_n \int_{t_0}^{t_n} dt_{n-1} \cdots \int_{t_0}^{t_2} dt_1\, \hat{H}(t_n)\cdots\hat{H}(t_1) = \frac{1}{n!}\int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_n\, \mathcal{T}\{\hat{H}(t_1)\cdots\hat{H}(t_n)\}. \tag{47}$$

You can prove this equality by using the Heaviside. We will not do it in full generality here, but just illustrate the idea for $n = 2$.

In the case $n = 2$, the equality in Eq. (47) reads

$$\int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1\, \hat{H}(t_2)\hat{H}(t_1) = \frac{1}{2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, \mathcal{T}\{\hat{H}(t_1)\hat{H}(t_2)\},$$

where the limits of integration are decoupled and the time-ordering operator ensures the operators are ordered correctly. To prove this, we will start with the left-hand side and use the Heaviside step function to decouple the integration limits:

$$\int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1\, \hat{H}(t_2)\hat{H}(t_1) = \int_{t_0}^{t} dt_2 \int_{t_0}^{t} dt_1\, \hat{H}(t_2)\hat{H}(t_1)\,\theta(t_2-t_1) \tag{48}$$

Compare this with the right-hand side using the definition of the time-ordering operator:

$$\begin{aligned}&\frac{1}{2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, \mathcal{T}\{\hat{H}(t_1)\hat{H}(t_2)\} \\ &\qquad = \frac{1}{2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, \left[\hat{H}(t_1)\hat{H}(t_2)\theta(t_1-t_2) + \hat{H}(t_2)\hat{H}(t_1)\theta(t_2-t_1)\right].\end{aligned} \tag{49}$$

Splitting this into two integrals and renaming the dummy integration variables $t_1 \leftrightarrow t_2$ in the first term, we find that Eq. (48) and Eq. (49) are equal.

Substituting Eq. (47) into Eq. (44) gives the Dyson series in its more familiar form:

$$\hat{U}(t,t_0) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_n \, \mathcal{T}\{\hat{H}(t_1)\cdots\hat{H}(t_n)\} \tag{50}$$

$$= \boxed{\mathcal{T} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} dt' \, \hat{H}(t')\right]}. \tag{51}$$

Note how we pulled the time-ordering operator outside the sum, since it acts on each term in the same way, and then recognized the Taylor expansion of the exponential function. This is sometimes referred to as the *time-ordered exponential*.